SUBJECT bt403' (past papers)

CR tasha KHAN

VU MEDICAL ZONE
BIOTECHONLOGISTS AND ZOOLOGISTS

ADMINS: Hafiza Mubeen
Muhammad Nouman
Tasha Khan
Zurraiz Butt
and amaan khan with vu medical zone

BT403

1)*Write the names of the minerals added in basal media (2)*

Ans:The minerals consist of macronutrients such as nitrogen, potassium, phosphorus, calcium, magnesium, and sulfur, and micronutrients such as iron, manganese, zinc, boron, copper, molybdenum, and cobalt.

2)What are synthetic seeds(2)

Ans: A synthetic or artificial seed has been defined as a somatic embryo encapsulated inside a coating and is considered to be analogous to a zygotic seed. There are several different types of synthetic seed: 1. Somatic embryos encapsulated in a water gel, 2. Dried and coated somatic embryos, 3. Dried and uncoated somatic embryos, 4. Somatic embryos suspended in a fluid carrier, 5. Shoot buds encapsulated in a water gel.

Created by Universal Document Converter

3)What are the advantages of agrobacterium transformation (3)

Ans: Advantages of Agrobacterium- mediated transformation:

This is a natural method of gene transfer. Agrobacterium can conveniently infect any explant (cells/tissues/organs).

Even large fragments of DNA can be efficiently transferred. IV. Stability of transferred DNA is reasonably good. V. Transformed plants can be regenerated effectively cure

4)Write the names of invitro cell cultures (3)

An: Types of in vitro cultures 1. Callus Cultures 2. Cell Suspension Culture 3. Anther/Microspore Culture 4. Protoplast Culture 5. Embryo Culture 6. Meristem Culture

5)Write the names of invitro cell cultures ?)

Ans:Cell Suspension Culture 3. Anther/Microspore Culture 4. Protoplast Culture 5. Embryo Culture 6. Meristem Culture

6)Write the names of the minerals added in basal media (2)

Ans:The minerals consist of macronutrients such as nitrogen, potassium, phosphorus, calcium, magnesium, and sulfur, and micronutrients such as iron, manganese, zinc, boron, copper, molybdenum, and cobalt.

8)What are synthetic seeds(2)

Ans: A synthetic or artificial seed has been defined as a somatic embryo encapsulated inside a coating and is considered to be analogous to a zygotic seed. There are several different types of synthetic seed: 1. Somatic embryos encapsulated in a water gel, 2. Dried and coated somatic embryos, 3. Dried and uncoated somatic embryos, 4. Somatic embryos suspended in a fluid carrier, 5. Shoot buds encapsulated in a water gel.

10)What are the advantages of agrobacterium transformation (3) Ans:Advantages of Agrobacterium-mediated transformation:

This is a natural method of gene transfer. Agrobacterium can conveniently infect any explant (cells/tissues/organs).

Even large fragments of DNA can be efficiently transferred. IV. S-+tability of transferred DNA is reasonably good. V. Transformed plants can be regenerated effectively cure

11)Write the names of invitro cell cultures (3)

ANS: Types of in vitro cultures 1. Callus Cultures 2. Cell Suspension Culture 3. Anther/Microspore Culture 4. Protoplast Culture 5. Embryo Culture 6. Meristem Culture

12)Discuss the surface sterilization of explant (5)

ANS: Surface Sterilization of Explants

Plant tissues inherently have various bacteria and fungi on their surfaces.

It is important that the explant be devoid of any surface contaminants prior to tissue culture since contaminants can grow in the culture medium, rendering the culture nonsterile.

Created by Universal Document Converter

dian fra In addition, they compete with the plant tissue for nutrition, thus depriving the plant tissue of nutrients.

Bacteria and especially fungi can rapidly overtake plant tissues and kill them.

The surface sterilants chosen for an experiment typically depend on the type of explant and also plant species.

Explants are commonly surface-sterilized using sodium hypochlorite

(household bleach), ethanol, and fungicides when using field-grown tissues.

The time of sterilization is dependent on the type of tissue; for example, leaf tissue will require a shorter sterilization time than will seeds with a tough seed coat.

Wetting agents such as Tween added to the sterilant can improve surface contact with the tissue.

Although surface contamination can be eliminated by sterilization, it is very difficult to remove contaminants that are present inside the explant that may show up at a later stage in culture.

This internal contamination can be controlled to a certain extent by frequent transfer to fresh medium or by the use of a low concentration of antibiotics in the medium.

Overexposing tissues to decontaminating chemicals can also kill tissues, so there is a balancing act between sterilizing explants and killing the explants themselves.

13)Three methods of micro propagation? (2,3?

ANS:There are three methods used for micropropagation: 1. Enhancing axillary bud breaking, 2. production of adventitious buds, 3. Somatic embryogenesis.

14)What is somatic hybridization.(5)

ANS:Somatic Hybridization

Protoplast fusion and somatic hybridization techniques provide the opportunity for bypassing reproductive isolation barriers, thus facilitating gene flow between species.

Fusion of protoplasts is accomplished by the use of PEG (polyethylene glycol].

Protoplast fusion has helped in the development of somatic hybrids or cybrids (cytoplasmic hybrids).

Protoplasts offer the possibility of efficient and direct gene transfer to plant cells. O DNA uptake has been found to be easier in protoplasts than into intact plant cells. Although protoplasts seem to be a very attractive means for plant regeneration and gene transfer, they are very vulnerable to handling.

One has to be very careful when manipulating protoplasts.

They have to be cultured on a medium with a high osmoticum such as sucrose or mannitol; otherwise the protoplasts will burst open, which is why plant regeneration from protoplasts has proven to be difficult.

Therefore, protoplasts are now used in cell culture studies mostly to study localization of proteins and transient transgene assays.

15)WHAT IS Importance of hairy roots? (2,3)

Created by Universal Document Converter

ANS:Importance of hairy roots:

Hairy roots can be cultured in vitro, and thus are important in plant biotechnology.

Hairy root systems are useful for the production of secondary metabolites, particularly pharmaceutical proteins.

. Transformed plants can be regenerated effectively.

16)what are Limitations of Agrobacterium-mediated transformation ?

ANS:There is a limitation of host plants for Agrobacterium, since many crop plants (monocotyledons e.g. cereals) are not infected by it. In recent years, virulent strains of

Agrobacterium that can infect a wide range of plants have been developed. II. The cells that regenerate more efficiently are often difficult to transform, e.g.

embryonic cells lie in deep layers which are not easy targets for Agrobacterium.

17). What is somaclonal variations? 2 marks

Ans: The genetic variations found in the in vitro cultured cells are collectively referred to as somaclonal variations. 2. How herbicides resistant crops are produced? 2 marks Ans: How to produce herbicide resistant crops

A number of biological manipulations involved in genetic engineering are in use to develop herbicide resistance plant

• Over expression of EPSPS gene • Use of mutant EPSPS gene • Detoxification of herbicide by a foreign gene

18)What is germplasm preservation? 3 marks

Ans: • Germplasm is living tissue from which new plants can be grown. It can be a seed or another plant part a leaf, a piece of stem, pollen or even just a few cells that can be turned into a whole plant. • Germplasm contains the information for a species' genetic makeup, a valuable natural resource of plant diversity 4. Why synthetic seed are produced? 3 marks Ans: • In some of the horticultural crops seeds propagation is not successful due to; • Heterozygosity of seeds particularly in cross pollinated crops • Minute seed size eg; orchids • Presence of reduced endosperm • Some seeds require mycorrhizal fungi association for germination eg: orchids • No seeds are formed

19). Write a note on protoplast culture 5 marks

Ans: Protoplast Culture

• Isolated protoplast can be cultured in an appropriate medium to reform cell wall and generate callus • Optimal culture conditions:

1. Optimal density to the culture.

2. Optimal auxin to cytokinin ratio, glucose and sucrose.

3. Maintain osmoprotectant in the medium

4. Temperature: 20-28°C pH: 5.5-5.9 0.25% Casein hydrolysate BAP and NAA

Created by Universal Document Converter

20)what is Culture of protoplasts ?

ANS:Protoplasts cultured in suitable nutrient media first generate a new cell wall • The formation of a complete cell with a wall is followed by an increase in size, number of cell organelles, and induction of cell division • The first cell division may occur within 2 to 7 days of culture • Resulting in small clumps of cell, also known as micro colony, within 1 to 3 weeks

From such clumps, there are two routes to generate a complete plant (depending on the species)

1. Plants are regenerated through organogenesis from callus masses

2. The micro calli can be made to develop into somatic embryos, which are then converted into whole plant through germination

21). Write a a note on contact herbicides and selected herbicides. 5 marks

Ans: • Contact herbicides: Contact is a word that means the chemical in that specific type of herbicide will kill the parts of the plant it contacts.

• For broadleaf weeds this means it will kill the above ground leafy part of the plants.

• It will not directly kill the below ground plants parts, such as roots, bulbs, tubers, or rhizomes.

• Contact herbicides are popular because they work quickly by killing the tissue in as fast as one day.

• Some herbicides will combine contact with systemic chemicals for a faster effect. • New Round Up Weed and Grass Killer has combined both for faster control.

• For systemic types of herbicides, the word "Systemic" means the plant absorbs through the leaves or stems and transports it internally throughout the plant. • The chemical travels with the sap so it usually doesn't have the quick "knockdown" effect. • The greatest benefit of a systemic type of herbicide is that it will kill the entire plant, roots and all. • The speed of chemical movement in the plant is largely dependent on soil and air temperature.

A chemical sprayed in early spring may take a couple weeks longer to work than the same chemical sprayed in mid-summer.

• The speed of kill is also dependent on the "mode of action" of the chemical (how the chemical works inside the plant).

22).what agriculture biotechnology give services to fuel the world? 2 marks

Ans:

Fuel the world

• Using biofuels

• Improving manufacturing process efficiency

• Reducing use of and reliance on petrochemicals

23).what is importance of protoplast culture 3 marks

Created by Universal Document Converter

Ans: • The protoplast in culture can be regenerated into a whole plant. • Hybrids can be developed from protoplast fusion. • It is easy to perform single cell cloning with protoplasts. • Genetic transformations can be achieved through genetic engineering of protoplast DNA. • Protoplasts are excellent materials for ultra-structural studies.

• Isolation of cell organelles and chromosomes is easy from protoplasts. • Protoplasts are useful for membrane studies (transport and uptake processes). • Isolation of mutants from protoplast cultures is easy.

24).why mature embryo need to culture 2 makrs

Ans: • Mature embryos are isolated from ripe seeds and cultured in vitro. Mature embryo cultures are carried out in the following conditions

conditions

1. When the embryos remain dormant for long periods.

2. Low survival of embryos in vivo.

3. To avoid inhibition in the seed for germination.

4. For converting sterile seeds to viable seedlings

25). what are competent cells 22 marks

Ans: • Competent Cells are more likely to incorporate foreign DNA if their cell walls are altered so that DNA can pass through more easily. • Such cells are said to be competent

26.why lettuce favored over tobacco ? 3marks

Ans: • Most of the plant is leaf tissue and this tissue contains the greatest number of plastids per cell • Unlike tobacco, lettuce has no toxic alkaloids that need to be removed - low purification and downstream processing costs • Lettuce is a relevant human foodstuff that can be consumed without cooking

27). What is post mitotic stage of anther? 3 marks

Ans: • Early bi-cellular stage of pollen development is most suitable for culture in Atropa belladonna and Nicotiana sp. etc. • Mature anthers are usually unsuitable for culture, but in Brassica oleracea mature anthers are the proper stage for

anther culture. • Anthers of proper stage are chosen by selecting flower buds of definite length under fixed environmental conditions.

28). What is Primary benefits of pesticides.

Ans: Controlling pests and plant disease vectors • Improved crop/livestock yields

• Improved crop/livestock quality • Invasive species controlled

29). What is short term effect of pesticides 5 marks

Ans: • Short-term pesticide poisoning or acute toxicity from pesticides is usually the result of a single and brief exposure to a pesticide.

• This kind of poisoning can happen due to exposure via the skin, inhalation, through the eyes, or orally.

Created by Universal Document Converter

• Symptoms of acute toxicity can become apparent instantly or take as long as 48 hours

30). What is name basic techniques of tissue culture? 5 mars

Ans: Culture vessels

• Cultures can be grown in various kinds of vessels such as petri plates, test tubes, “Magenta boxes,” bottles, and flasks

Culture medium

• The important media used for all purpose experiment are Murashige and Skoog medium (MS medium) • The culture medium is closed with cotton plug/ or aluminium foil sheet. • The pH of the medium is adjusted to 5.8 (acidic range).

Composition of culture medium

• Nutrient Medium • Medium depends upon the type of plant tissue or cell used for culture • Generally nutrient consist of • inorganic salts (both micro & macro elements) a carbon source (usually sucrose) • Vitamins (eg. nicotinic acid,

thiamine, pyridoxine • Amino acids (eg. arginine) • Growth regulators (eg. auxins) • An optimum pH (5.7) is also vary important

Sterilization

• Sterilization Methods Used in Tissue Culture Laboratory

• All the materials, e.g., vessels, instruments, medium, plant material, etc., used in culture work must be freed from microbes

Sterilization techniques

• sterilization is achieved by one of the following approaches: • dry heat treatment • flame sterilization • autoclaving • filter sterilization • wiping with 70% ethanol • surface sterilization.

Inoculation

• Transfer of explant (root, stem, leaf, etc.) on to a culture medium is called inoculation. • The inoculation is carried out under aseptic condition for which an apparatus called laminar air flow chamber is used. • Flamed and cooled forceps are used for transfer of plant materials to different culture media kept in glasswares.

Incubation

• The culture medium with the inoculum is incubated at 26 - 28oC with the light intensity at 2000 to 4000 lux (unit of intensity of light) and allowing photoperiod of 16 hour of light and 8 hours of darkness.

Induction of callus

• Due to activity of auxins and cytokinins, the explant is induced to form callus. • The callus is an unorganized mass of undifferentiated tissue. • The mechanism of

callus formation is that auxin induce cell elongation and cytokinin induces cell division as a result of which masses of cells are formed.

Created by Universal Document Converter

Morphogenesis

• Formation of new organs from the callus under the influence of auxin and cytokinin is called morphogenesis. • Roots and shoots are differentiated from the callus. • Such embryos are called somatic embryos result in the formation of young plantlet.

Types of morphogenesis

• Organogenesis • Embryogenesis

Organogenesis

• Formation of new organs such as shoot and root is known as organogenesis. • The development of shoot from the callus is called caulogenesis and formation of root is called rhizogenesis respectively.

Embryogenesis

• Formation of embryos (ie. bipolar structure having shoot and root) from the callus is called embryogenesis. • These embryos arise from somatic callus tissue and are called somatic embryos or embryoids or somaclonal embryos.

Hardening

• Exposing the plantlets to the natural environment in a stepwise manner is known as hardening. • Finally the plantlets are gradually transferred to the soil.

31) what is somoclonal variations and germinal variations?

Ans: • Somaclonal variation is the variation seen in plants that have been produced by plant tissue culture. Chromosomal rearrangements are an important source of this variation • Gametoclonal variation has been defined as the variation among plants regenerated from gametic cells in culture

32).What are the importance of biotechnology Vecine ?

Ans: Vaccines

• Biotechnology-derived vaccines are used in livestock and humans.

• They may be cheaper, better and/or safer than traditional vaccines.

• They are also stable at room temperature, and do not need refrigerated storage

• Biotechnology-derived vaccines are used in livestock and humans. • Cheaper • better • safer than traditional vaccines • stable at room temperature • do not need refrigerated storage

33)How chloroplast is transmit write any 2 procedure

Ans: Biolistic Transfer:

• Biolystic is one of the efficient approaches for plastid transformation. • Effective penetration and high transfer frequency are some of the plus points of biolystic method. • There are number of bacteria and viruses are known to infect chloroplast.

Created by Universal Document Converter

• It's envelop is made up of double membrane and actually considered that chloroplast transformation was considered to be virtually impossible. • However, invention of biolistic gene gun technology paved the pathway of direct delivery of target gene into the living cell.

• It is fortunate that DNA is deposited in a chloroplast and successfully integrated into the chloroplast genome.

34)what is PEG Mediated Transformation: ?

ANS:• Polyethylene glycol is widely used in transformation work. Despite its efficiency, PEG mediated transformation is far behind than the biolistic approach. • Foreign DNA is taken by protoplast in presence of PEG and transported by unknown

process from cytoplasm into the chloroplast and finally integrated into the genome

35). How explant transfer to culture plate ?

Ans: Establishment of the explant in a culture medium

• The medium sustains the plant cells and encourages cell division.

• It can be solid or liquid

36) what is Protoplast culture ?

ANS:Total question 26 20 Mcqs: 1.A germplasm is a collection of genetic resources for an organism. 2.64% of total soya crop 3.Frdrc grfth exprmnt date---- 1928 4.Morphological marker ---- are those traits that are scored visually, or morphological markers are those genetic markers whose inheritance can be followed with the naked eye 5.Herbicide type systematic type 6.In which the culture can be prepared...ans all above 7.Drought is .....Stomata closure. 8.Anthers or pollens can be cultured on a suitable medium containing sucrose (usually 2%), 9. Surface sterilize with 0.01% Tween-20 for 15 min 10. The process of growing an entire plant from a single cell or group of cells. Regeneration 11. Transposable elements (TEs), also known as "jumping genes" or transposons, 12. particles of gold or tungsten 13.Photosynthesis is carried out in presence if which plastid....chloroplast 14.What is not the effect of mild pesticide? 15. growth is change in .....size 16 .... Is used to kill the mite and insects eggs.

37) What are herbicides?

Ans: A herbicide is a chemical substance used to control or manipulate undesirable vegetation, especially weeds.

• Herbicides are extensively used in gardening, farming, and landscape turf management.

• Herbicides tend to have wide-ranging effects on non-target species (other than those the pesticide is meant to control or kill).

38)What are environmental effects of pesticides? 5marks question

Ans: Environmental effects

• Widespread use of pesticides in agriculture has experts worried due to their longterm environmental damage.

Created by Universal Document Converter

• Some pesticides can stick around for years, posing a very real threat to the ecological system and hence human health.

• Excessive and careless use of pesticides can contaminate water sources and soil, make fruits and vegetables less nutritious, and reduce biodiversity.

• Some pesticides have also been linked to the dramatic reduction in the number of bees across the world, posing a huge threat to agriculture and food security, given that bees pollinate more than 70% of all crops.

39) what are Benefits of agri biotechnology?

Ans: Benefits of Biotechnology

• HEAL THE WORLD

• FUEL THE WORLD

• FEED THE WORLD

Heal the world

• Creating more precise tools for disease detection

• Reducing rates of infectious disease

Fuel the world

• Using biofuels

• Improving manufacturing process efficiency

• Reducing use of and reliance on petrochemicals

Feed the world

• Generating higher crop yields

• Using biotech crops

Improving food and crop oil content

40) Why we need herbicides?

Ans: • Weeds are unwanted & useless plants that grow along with the crop plants . • Weeds compete with the crops for light & nutrients, besides harboring various pathogens . • So it is estimated that the worlds crop yield is reduced by 10 – 15 % due to the presence of weeds. • To tackle the problem of weeds , modern agriculture has developed a wide range of weed killers (herbicides).

• Herbicides are broad spectrum as they can kill wide range of weeds.

mark

41)what are Characteristics of synthetic seeds. 3

Ans: • High volume. Large scale propagation method • Maintains genetic uniformity of plants • Direct delivery of propagules to the field, thus eliminating transplants • Lower cost per plantlet • Rapid multiplication of plants

Created by Universal Document Converter

42) what are Limitation of PEG 3

Ans: 1. The DNA is susceptible for degradation and rearrangement. 2. Random integration of foreign DNA into genome may result in undesirable traits. 3. Regeneration of plants from transformed protoplasts is a difficult task. Why biotechnological vaccines are important? 5 Write the methods of chloroplast transformation. 5

43) How pesticides effect the kids more as compared to adults? (2 marks)

Ans• Children are especially susceptible to harmful effects of pesticides.

• They can easily become exposed to pesticides (via inhalation or skin contact) in schools, daycare, playgrounds, hospitals, and any other public areas, no matter how careful you are

44) How viruses sepread from plant to plant? (3 marks)

Ans: Viruses can be spread from plant to plant by several means. Some of these would include transmission from the parent plant to an offspring through the genetic structure of the plants. • Other ways in which viruses can be transmitted

are through vegetative propagation, grafting and budding, seed transmission and mechanical spread by insects and man

45) what are Factor of herbicides?

Ans: • Because weeds contain a tremendous amount of genetic variation that allows them to survive under a variety of environmental conditions the development of a resistant species is brought about through selection pressure imposed by the continuous use of an herbicide.

• Factors that can lead to or accelerate the development of herbicide resistance include

• weed characteristics • Herbicide characteristics • and cultural practices.

46)Microinjecti n ?

Ans: • where the DNA is directly injected into plant protoplasts or cells (specifically into the nucleus or cytoplasm) using fine tipped (0.5 - 1.0 micrometerdiameter) glass needle or micropipette. • This method of gene transfer is used to introduce DNA into large cells such as oocytes, eggs, and the cells of early embryo.

47.What are the importance of biotechnology Vecine

Ans: Vaccines

• Biotechnology-derived vaccines are used in livestock and humans.

• They may be cheaper, better and/or safer than traditional vaccines.

• They are also stable at room temperature, and do not need refrigerated storage

• Biotechnology-derived vaccines are used in livestock and humans. • Cheaper • better • safer than traditional vaccines • stable at room temperature • do not need refrigerated storage .

48) How chloroplast is transmit write any 2 procedure

Ans: Biolistic Transfer:

Created by Universal Document Converter

• Biolystic is one of the efficient approaches for plastid transformation. • Effective penetration and high transfer frequency are some of the plus points of biolystic method. • There are number of bacteria and viruses are known to infect chloroplast.

• It's envelop is made up of double membrane and actually considered that chloroplast transformation was considered to be virtually impossible. • However, invention of biolistic gene gun technology paved the pathway of direct delivery of target gene into the living cell.

• It is fortunate that DNA is deposited in a chloroplast and successfully integrated into the chloroplast genome.

49How explant transfer to culture plate?

Ans: Establishment of the explant in a culture medium

• The medium sustains the plant cells and encourages cell division.

• It can be solid or liquid

50). Protoplast culture

Total question 26 20 Mcqs: 1.A germplasm is a collection of genetic resources for an organism. 2.64% of total soya crop 3.Frdrc grfth exprmnt date---- 1928 4.Morphological marker ---- are those traits that are scored visually, or morphological markers are those genetic markers whose inheritance can be followed with the naked eye 5.Herbicide type systematic type 6.In which the culture can be prepared...ans all above 7.Drought is .....Stomata closure. 8.Anthers or pollens can be cultured on a suitable medium containing sucrose (usually 2%), 9. Surface sterilize with 0.01% Tween-20 for 15 min 10. The process of growing an entire plant from a single cell or group of cells. Regeneration 11. Transposable elements (TEs), also known as "jumping genes" or transposons, 12. particles of gold or tungsten 13.Photosynthesis is carried out in presence if which plastid....chloroplast 14.What is not the effect of mild pesticide? 15. growth is change in .....size 16 .... Is used to kill the mite and insects eggs.

50)Why we need herbicides?

Ans: • Weeds are unwanted & useless plants that grow along with the crop plants . • Weeds compete with the crops for light & nutrients, besides harboring various pathogens . • So it is estimated that the worlds crop yield is reduced by 10 – 15 % due to the presence of weeds. • To tackle the problem of weeds , modern agriculture has developed a wide range of weed killers (herbicides).

• Herbicides are broad spectrum as they can kill wide range of weeds.

51)what are Characteristics of synthetic seeds.

Ans: • High volume. Large scale propagation method • Maintains genetic uniformity of plants • Direct delivery of propagules to the field, thus eliminating transplants • Lower cost per plantlet • Rapid multiplication of plants

52. How pesticides effect the kids more as compared to adults? (2 marks)

Ans• Children are especially susceptible to harmful effects of pesticides.

Created by Universal Document Converter

• They can easily become exposed to pesticides (via inhalation or skin contact) in schools, daycare, playgrounds, hospitals, and any other public areas, no matter how careful you are

53. How viruses sepread from plant to plant? (3 marks)

Ans: Viruses can be spread from plant to plant by several means. Some of these would include transmission from the parent plant to an offspring through the genetic structure of the plants. • Other ways in which viruses can be transmitted

are through vegetative propagation, grafting and budding, seed transmission and mechanical spread by insects and man. 4. How synthetic seeds are produced? (3 marks)

54) what are Factor of herbicides?

Ans: • Because weeds contain a tremendous amount of genetic variation that allows them to survive under a variety of environmental conditions the development of a resistant species is brought about through selection pressure imposed by the continuous use of an herbicide.

• Factors that can lead to or accelerate the development of herbicide resistance include

• weed characteristics • Herbicide characteristics • and cultural practices.

55) what is Culture protoplast?

ANS:A cell without a cell wall is called a protoplast. An earlier definition of protoplast (cells with their walls stripped off and removed from the proximity of their neighbouring cells) was modified by Vasil:

The protoplast is a part of **plant** cell which lies within the cell wall and can be plasmolysed and isolated by removing the cell wall by mechanical or enzymatic procedure ***[170]***.

Major developments in **protoplast culture** are highlighted in Table 2.4. Somatic hybridization involves isolation, fusion, selection, and culture of the hybrid cells for the regeneration of hybrid **plants** from them.

56)What is germplasm preservation? 3 marks

Ans: • Germplasm is living tissue from which new plants can be grown. It can be a seed or another plant part a leaf, a piece of stem, pollen or even just a few cells that can be turned into a whole plant. • Germplasm contains the information for a species' genetic makeup, a valuable natural resource of plant diversity 4. Why synthetic seed are produced? 3 marks Ans: • In some of the horticultural crops seeds propagation is not successful due to; • Heterozygosity of seeds particularly in cross pollinated crops • Minute seed size eg; orchids • Presence of reduced endosperm • Some seeds require mycorrhizal fungi association for germination eg: orchids • No seeds are formed

57) Write a note on protoplast culture 5 marks

Ans: Protoplast Culture

• Isolated protoplast can be cultured in an appropriate medium to reform cell wall and generate callus • Optimal culture conditions:

1. Optimal density to the culture.

2. Optimal auxin to cytokinin ratio, glucose and sucrose.

Created by Universal Document Converter

3. Maintain osmoprotectant in the medium

4. Temperature: 20-28°C pH: 5.5-5.9 0.25% Casein hydrolysate BAP and NAA

57) Write a a note on contact herbicides and selected herbicides. 5 marks

Ans: • Contact herbicides: Contact is a word that means the chemical in that specific type of herbicide will kill the parts of the plant it contacts.

• For broadleaf weeds this means it will kill the above ground leafy part of the plants.

• It will not directly kill the below ground plants parts, such as roots, bulbs, tubers, or rhizomes.

• Contact herbicides are popular because they work quickly by killing the tissue in as fast as one day.

• Some herbicides will combine contact with systemic chemicals for a faster effect. • New Round Up Weed and Grass Killer has combined both for faster control.

• For systemic types of herbicides, the word "Systemic" means the plant absorbs through the leaves or stems and transports it internally throughout the plant. • The chemical travels with the sap so it usually doesn't have the quick "knockdown" effect. • The greatest benefit of a systemic type of herbicide is that it will kill the entire plant, roots and all. • The speed of chemical movement in the plant is largely dependent on soil and air temperature.

A chemical sprayed in early spring may take a couple weeks longer to work than the same chemical sprayed in mid-summer.

• The speed of kill is also dependent on the "mode of action" of the chemical (how the chemical works inside the plant).

58.what agriculture biotechnology give services to fuel the world? 2 marks

Ans: Fuel the world

• Using biofuels • Improving manufacturing process efficiency

• Reducing use of and reliance on petrochemicals

59) what are competent cells 2 marks

Ans: • Competent Cells are more likely to incorporate foreign DNA if their cell walls are altered so that DNA can pass through more easily. • Such cells are said to be competent

60)What is genetic marker?

A genetic marker is a gene or DNA sequence with a known location on a chromosome that can be used to identify individuals or species. It can be described as a variation (which may arise due to mutation or alteration in the genomic loci) that can be observed

61)What are synthetic seeds(2)

. Synthetic seeds are defined as artificially encapsulated somatic embryos, shoot buds, cell aggregates, or any other tissue that can be used for sowing as a seed and that possess the ability to convert into a plant under in vitro or ex vitro conditions and that retain this potential also after storage.

Created by Universal Document Converter